

Dataset found at <https://divvybikes.com/>

**How do annual  
subscribers & casual  
customers use the  
bikes differently?**

**Divvy Bike Share, Chicago**

Python analysis by Elodie Barthout ([Github](#))

*Last update: June 22nd, 2026*

# Dataset description

- Divvy is a **bike-share app** from the city of Chicago
- Analysis covering the years **2018-2019** (*8 datasets merged together to get a big picture*)
- Dataset contains, for each ride:
  - Date & time
  - Start & end stations (IDs & names)
  - User type (Subscriber or Customer)
- Limitation: **no user IDs** so we can't analyze user growth or determine the number of people subscribing to the service compared to the casual riders.

# Analysis of bike rentals over the years 2018–2019

More than **6 million of rides** over this time period (6 295 548 entries):

**93% done by subscribers** (*annual memberships*)

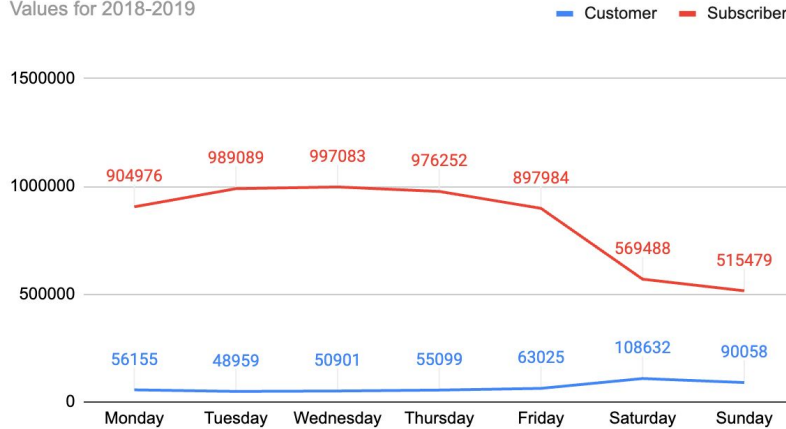
**7% done by casual customers** (*single ride or day pass users*)

→ Majority of users being older than 30 years-old (59%)

# Differences between subscribers & customers

Number of daily bike rides

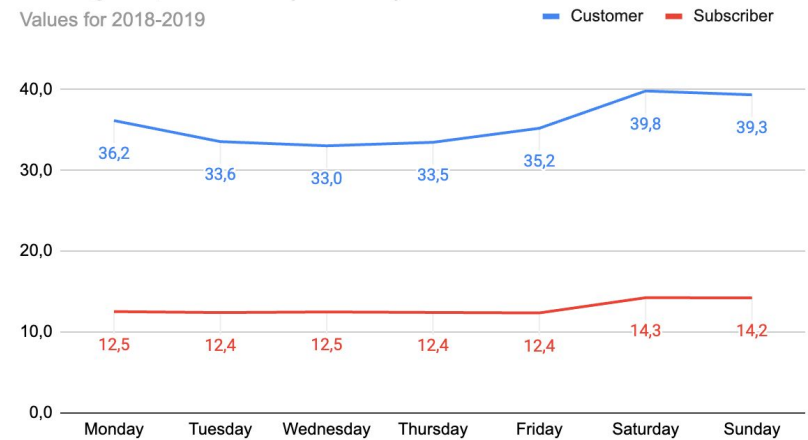
Values for 2018-2019



Subscribers mainly use the service during the week while customers seem to use it more during weekends

Average trip duration (minutes)

Values for 2018-2019



On average, customers take longer trips than subscribers (2-3 times longer), while both ride longer during weekends

# Do customers take round trips?

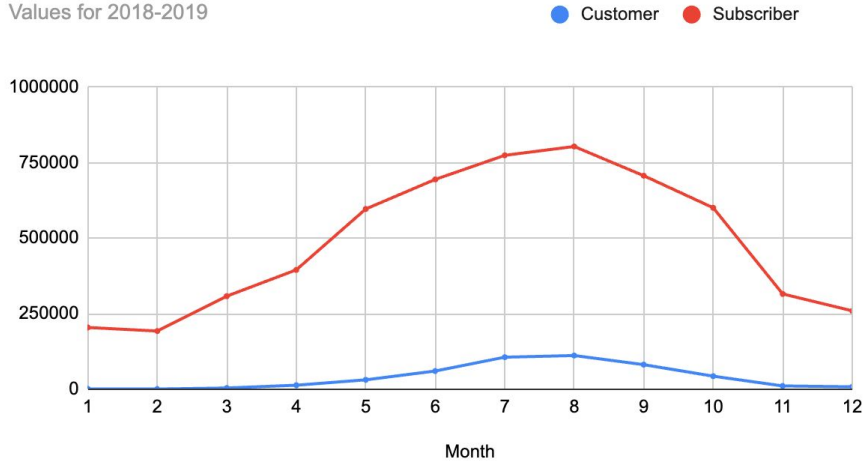
|            | From                         | To                           | Top |
|------------|------------------------------|------------------------------|-----|
| Subscriber | Canal St & Adams St          | Michigan Ave & Washington St | 1   |
|            | Michigan Ave & Washington St | Canal St & Adams St          | 2   |
|            | Michigan Ave & Washington St | Clinton St & Washington Blvd | 3   |
|            | Columbus Dr & Randolph St    | Clinton St & Washington Blvd | 4   |
|            | Wacker Dr & Washington St    | Michigan Ave & Washington St | 5   |
| Customer   | Streeter Dr & Grand Ave      | Streeter Dr & Grand Ave      | 1   |
|            | Lake Shore Dr & Monroe St    | Streeter Dr & Grand Ave      | 2   |
|            | Lake Shore Dr & Monroe St    | Lake Shore Dr & Monroe St    | 3   |
|            | Michigan Ave & Oak St        | Michigan Ave & Oak St        | 4   |
|            | Streeter Dr & Grand Ave      | Millennium Park              | 5   |

The most popular start & end stations for casual customers are the same stations. Further analysis showed that **8,3% of casual rides were round trips**, compared to 1,6% for subscribers. This suggests that casual customers are more likely to use bikes for recreational or sightseeing trips rather than point-to-point transportation. This is reinforced by the shore localization of the most popular stations used by casual riders.

# Are there more profitable months?

## Number of monthly rides

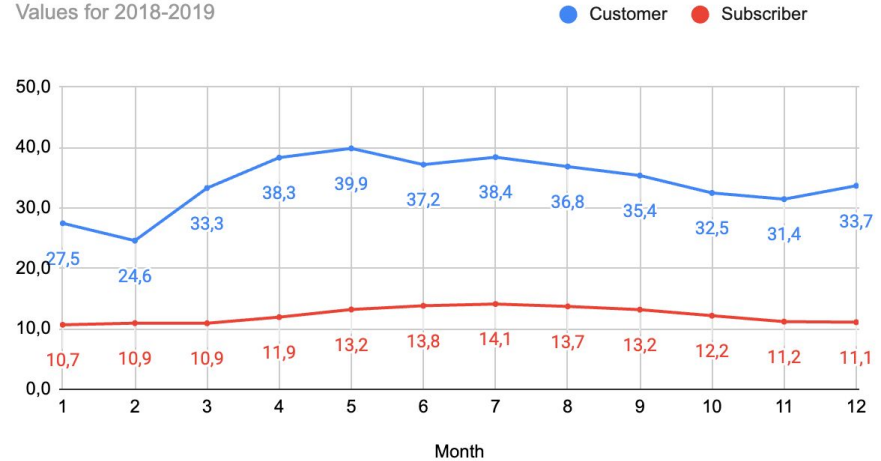
Values for 2018-2019



**July & August** are the months with the highest number of rides, regardless of the user type

## Average trip duration (minutes)

Values for 2018-2019



Trip duration slightly increasing during Spring & Summer, and being the lowest during Winter

# Conclusion

- Subscribers mostly use the service during the week for an average point-to-point trip of 12 min
- Customers mostly use the service during the weekend for an average 39 min trip, mainly taking round trips around sightseeing points
- July & August are the most profitable months and represent nearly 30% of the rides

→ Based on these results, it is fair to assume that customers are visitors renting bikes for leisure while subscribers use the bikes to commute to work.